

Solve multi-step measure problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut? Copy or complete the first step.

2. Convert 4.375 km to metres.

3. Convert 2.65 kg to grams and 3,750 ml to litres.

4. Miro says: Miro multiplies whenever converting, regardless of the direction between units. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut? Copy or complete the first step.
Answer 31 complete pieces, with 10 cm left.
2. Convert 4.375 km to metres.
Answer 4,375 m.
3. Convert 2.65 kg to grams and 3,750 ml to litres.
Answer 2,650 g and 3.75 litres.
4. Miro says: Miro multiplies whenever converting, regardless of the direction between units. Is this correct? Explain.
Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.

Solve multi-step measure problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

6. Check this result using an inverse, estimate or known fact: Convert 2.65 kg to grams and 3,750 ml to litres.

2. Convert 4.375 km to metres.

7. Prove or explain your reasoning: Write 845 mm in metres.

3. Convert 2.65 kg to grams and 3,750 ml to litres.

8. Identify and correct this misconception: Miro multiplies whenever converting, regardless of the direction between units.

4. Solve this independently and show every stage: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Convert 4.375 km to metres.

10. Write and solve a similar question to: Convert 2.65 kg to grams and 3,750 ml to litres.

Teacher answers and checking prompts

1. A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
Answer 31 complete pieces, with 10 cm left.
2. Convert 4.375 km to metres.
Answer 4,375 m.
3. Convert 2.65 kg to grams and 3,750 ml to litres.
Answer 2,650 g and 3.75 litres.
4. Solve this independently and show every stage: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
Answer 31 complete pieces, with 10 cm left.
5. Use a second method or representation for: Convert 4.375 km to metres.
Answer Expected result: 4,375 m. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Convert 2.65 kg to grams and 3,750 ml to litres.
Answer Expected result: 2,650 g and 3.75 litres. A valid check must be shown.
7. Prove or explain your reasoning: Write 845 mm in metres.
Answer 0.845 m. The explanation must show why the method works.
8. Identify and correct this misconception: Miro multiplies whenever converting, regardless of the direction between units.
Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Convert 2.65 kg to grams and 3,750 ml to litres.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve multi-step measure problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Write 845 mm in metres.

6. Create a new example that tests the same idea as: Convert 2.65 kg to grams and 3,750 ml to litres.

2. Prove the answer to this question, rather than only calculating it: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Convert 4.375 km to metres.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 2,650 g and 3.75 litres.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro multiplies whenever converting, regardless of the direction between units.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Write 845 mm in metres.
Answer 0.845 m. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A 12.5 m roll is cut into 40 cm pieces. How many complete pieces can be cut?
Answer Expected result: 31 complete pieces, with 10 cm left. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Convert 4.375 km to metres.
Answer Expected result: 4,375 m. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 2,650 g and 3.75 litres.
Answer One valid reconstruction is: Convert 2.65 kg to grams and 3,750 ml to litres. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro multiplies whenever converting, regardless of the direction between units.
Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.
6. Create a new example that tests the same idea as: Convert 2.65 kg to grams and 3,750 ml to litres.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.